

An Exact Cooperation Formula for Introspection Dynamics in the Heterogeneous Public Goods Game

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Abstract

We study introspection dynamics on multiplayer games in which the payoff difference Δf_i a player evaluates when considering a strategy switch is independent of all other players' current actions, a property we call *state-independence*. Under this condition the introspection Markov chain decomposes as a random-scan product of N independent two-state chains, one per player, and the stationary distribution is a product measure. As our main application we consider the heterogeneous public goods game, where N players may differ in their contributions α_i , public goods multipliers r_i , and choice intensities β_i . We prove that the linear payoff structure implies state-independence, and the long-run cooperation probability admits the exact closed form

$$p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{1 - \mu_{i0} - \mu_{i1}}{1 + e^{\beta_i \alpha_i (1 - r_i/N)}} + \mu_{i0} \right],$$

with no asymptotic approximation. Several structural consequences follow immediately: a player-specific cooperation threshold at $r_i = N$ (under symmetric mutation), payoff-neutrality under zero choice intensity, and the sign of each player's sensitivity to their own parameters.

1 Introduction

The public goods game is a canonical model of cooperation in evolutionary game theory: players decide whether to contribute to a common resource, which is multiplied and redistributed equally, creating a tension between individual incentive and collective benefit [5]. Classical analyses assume a homogeneous population, but real groups are heterogeneous: players differ in their productivity, their contributions, and how sensitively they respond to payoff differences [4].

A natural learning rule for such settings is *introspection dynamics*, introduced in [1]: at each time step a player compares their current payoff to the payoff they *would have received* had they played the alternative action (keeping all other players fixed), and switches with a probability governed by this difference. Because the comparison is self-referential rather than imitative, introspection applies to arbitrary asymmetric games where role-model imitation is unavailable or unnatural.

In [2] the authors extended introspection dynamics to multiplayer asymmetric games and derived a formula for the stationary distribution that can be evaluated numerically for any game. In their general framework the transition rate for player i switching depends on the actions of all other players through the payoff function, so the stationary distribution must be computed by solving a linear system of dimension 2^N ; no closed form is available in general.

In this paper we show that whenever the payoff difference Δf_i a player evaluates when considering a switch is independent of all other players' current actions, a property we call *state-independence*, the introspection chain decomposes exactly as a product of N independent two-state chains (one per player), the stationary distribution is a product measure, and the cooperation probability p_C^{intro} admits an explicit closed form scaling as $O(N)$ rather than $O(2^N)$. We then show that the heterogeneous public goods game (PGG) satisfies state-independence as a consequence of the *linearity* of its payoff structure, yielding the exact formula (7).

The paper is organised as follows. Section 2 introduces the general framework and the state-independence condition. Section 3 proves, for any state-independent game, that individual cooperation probabilities are given by (3) and the stationary distribution is a product measure. Section 4 establishes state-independence for the PGG and derives the exact cooperation formula. Section 5 records several structural corollaries. Section 6 concludes.

2 Setup and notation

Consider N ordered players, each choosing from two actions $\{C, D\}$ coded as $\{1, 0\}$. The state space is $S = \{0, 1\}^N$ with $|S| = 2^N$. Each player i has a payoff function $f_i : S \rightarrow \mathbb{R}$ and player-specific choice intensity $\beta_i > 0$. Writing $\mathbf{a}_{i \rightarrow \bar{a}_i}$ for the state with player i 's action flipped to $\bar{a}_i = 1 - a_i$, the *payoff difference* for player i at state $\mathbf{a}$ is

$$\Delta f_i(\mathbf{a}) := f_i(\mathbf{a}) - f_i(\mathbf{a}_{i \rightarrow \bar{a}_i}).$$

A payoff function f_i is *state-independent* if $\Delta f_i(\mathbf{a})$ depends only on a_i , not on $\mathbf{a}_{-i}$. When this holds there is a constant $\delta_i \in \mathbb{R}$ such that

$$\Delta f_i(\mathbf{a}) = (1 - 2a_i) \delta_i, \tag{1}$$

where $\delta_i = \Delta f_i|_{a_i=0}$ is the payoff difference when player i defects. When $a_i = 0$ (1) follows from the definition of δ_i , when $a_i = 1$, we have: $\Delta f_i|_{a_i=1} = -\Delta f_i|_{a_i=0}$ which gives (1). To define introspection dynamics we introduce the following two quantities:

- **Fermi function.** Each player i has a personal Fermi function $\phi_i(x) = (1 + e^{\beta_i x})^{-1}$. Since $\beta_i > 0$, the exponential $e^{\beta_i x}$ is strictly increasing in x , so ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$ and $\phi_i(x) + \phi_i(-x) = 1$.
- **Mutation.** Each player i has mutation probabilities $\mu_{i0}, \mu_{i1} > 0$ with $\mu_{i0} + \mu_{i1} < 1$, where μ_{i0} is the probability of mutating to cooperation and μ_{i1} is the probability of mutating to defection, regardless of the intended choice of the dynamic. Having $\mu_{i0}, \mu_{i1} > 0$ ensures ergodicity of the Markov chain.

Introspection dynamics. At each step, one player i is selected uniformly at random. Player i considers switching from action a_i to the alternative $\bar{a}_i = 1 - a_i$. The transition probability for the switch is

$$T_{\mathbf{a}, \mathbf{a}_{i \rightarrow \bar{a}_i}}^{\text{intro}} = \frac{1}{N} [(1 - \mu_{i0} - \mu_{i1}) \phi_i(\Delta f_i(\mathbf{a})) + \mu_{i, \bar{a}_i}], \tag{2}$$

where $\mu_{i, \bar{a}_i}$ denotes μ_{i0} when $\bar{a}_i = C$ and μ_{i1} when $\bar{a}_i = D$. Since ϕ_i is strictly decreasing, a larger payoff gain from keeping the current action yields a smaller switching probability.

The main result of this paper is to give the overall cooperation probability for introspection dynamics for any *state-independent* payoff function.

Definition 1 (Cooperation probability). *Given an ergodic chain on S with stationary distribution π , the cooperation probability is $p_C = \pi \cdot s$, where $s_{\mathbf{a}} = \frac{1}{N} \sum_i a_i$ is the fraction of cooperators in state $\mathbf{a}$.*

3 Individual cooperation probabilities

For a state-independent game, $\Delta f_i(\mathbf{a})$ depends only on a_i , not on $\mathbf{a}_{-i}$. We use this to compute each player's long-run cooperation probability and the stationary distribution directly from the two-state update rule.

Theorem 2 (Individual cooperation probabilities). *For any state-independent game with constants δ_i , under introspection dynamics with player-specific choice intensity β_i and mutation probabilities $\mu_{i0}, \mu_{i1} > 0$ with $\mu_{i0} + \mu_{i1} < 1$, the long-run probability that player i cooperates is*

$$p_i = \phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0}. \quad (3)$$

Proof. By state-independence, $\Delta f_i(\mathbf{a}) = (1 - 2a_i)\delta_i$ depends only on a_i . When player i is selected, the probability their next action is C equals p_i regardless of their current action: if $a_i = D$, the switch probability is $\phi_i(\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i0} = p_i$; if $a_i = C$, the probability of remaining at C is $1 - [\phi_i(-\delta_i)(1 - \mu_{i0} - \mu_{i1}) + \mu_{i1}] = p_i$, using $\phi_i(x) + \phi_i(-x) = 1$. Since each selection of player i places them in state C with probability p_i independently of their current state, the long-run cooperation probability of player i is p_i . □

Theorem 3 (Stationary distribution). *For any state-independent game with constants δ_i , the stationary distribution of the introspection chain on $S = \{0, 1\}^N$ is the product measure*

$$\pi_{\mathbf{a}} = \prod_{i=1}^N p_i^{a_i} (1 - p_i)^{1-a_i}, \quad \mathbf{a} \in S, \quad (4)$$

where p_i is given by (3).

Proof. By Theorem 2, each selection of player i places them in state C with probability p_i independently of every other player's action. In stationarity the players' actions are therefore independent, with player i cooperating with probability p_i and defecting with probability $1 - p_i$. The probability of state $\mathbf{a}$ is the product of these marginals, giving (4). □

4 State-independence of the heterogeneous public goods game

The *heterogeneous public goods game* is played by N players with player-specific contributions $\alpha_1, \dots, \alpha_N > 0$ and public goods multipliers $r_1, \dots, r_N > 1$. The payoff to player i in state $\mathbf{a} = (a_1, \dots, a_N)$ is

$$f_i(\mathbf{a}) = \frac{r_i}{N} \sum_{j=1}^N \alpha_j a_j - \alpha_i a_i. \quad (5)$$

We write $C(\mathbf{a}) = \sum_j \alpha_j a_j$ for the total contribution. We now verify that this payoff is state-independent.

Lemma 4 (State-independence of the PGG). *The payoff (5) is state-independent for every player i , with*

$$\delta_i = \alpha_i \left(1 - \frac{r_i}{N}\right), \quad (6)$$

i.e. $\Delta f_i(\mathbf{a}) = (1 - 2a_i) \alpha_i (1 - r_i/N)$, independent of $\mathbf{a}_{-i}$.

Proof. When player i switches from a_i to $\bar{a}_i = 1 - a_i$, only their own contribution changes, so the total contribution becomes $C(\mathbf{a}_{i \rightarrow \bar{a}_i}) = C(\mathbf{a}) + \alpha_i(1 - 2a_i)$. Expanding both payoffs:

$$\begin{aligned} f_i(\mathbf{a}) &= \frac{r_i}{N} C(\mathbf{a}) - \alpha_i a_i, \\ f_i(\mathbf{a}_{i \rightarrow \bar{a}_i}) &= \frac{r_i}{N} [C(\mathbf{a}) + \alpha_i(1 - 2a_i)] - \alpha_i(1 - a_i). \end{aligned}$$

Subtracting:

$$\begin{aligned} \Delta f_i &= -\frac{r_i}{N} \alpha_i (1 - 2a_i) + \alpha_i (1 - 2a_i) \\ &= \alpha_i (1 - 2a_i) \left(1 - \frac{r_i}{N}\right). \end{aligned}$$

The term $C(\mathbf{a})$ cancels entirely, confirming state-independence with $\delta_i = \alpha_i(1 - r_i/N)$. $\square$

Corollary 5 (Exact cooperation probability). *For the heterogeneous public goods game (5), the long-run cooperation probability is*

$$p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N \left[\frac{1 - \mu_{i0} - \mu_{i1}}{1 + e^{\beta_i \alpha_i (1 - r_i/N)}} + \mu_{i0} \right]. \quad (7)$$

Proof. By Lemma 4, $\delta_i = \alpha_i(1 - r_i/N)$. By Theorem 2: $p_C^{\text{intro}} = \frac{1}{N} \sum_{i=1}^N p_i$. Substituting $\phi_i(\delta_i) = (1 + e^{\beta_i \alpha_i (1 - r_i/N)})^{-1}$ into (3) gives (7). $\square$

Figure 1 validates Corollary 5 against exact values and simulation across group sizes. Table 1 verifies Theorem 3: for $N = 3$ with per-player multipliers $r_1 = 1$, $r_2 = 3 = N$, $r_3 = 9$, uniform contributions $\alpha_i = 1$, $\beta = 2$, and $\mu_{i0} = \mu_{i1} = 0.1$, the individual cooperation probabilities are $p_1 \approx 0.267$, $p_2 = 0.500$, $p_3 \approx 0.886$, and the formula and exact values of $\pi_{\mathbf{a}}$ agree to five decimal places for all eight states.

5 Structural consequences

The formulas (3) and (7) allow immediate structural conclusions about individual and aggregate cooperation.

Corollary 6 (Player-specific cooperation thresholds). *Player i cooperates with probability greater than $\frac{1}{2}$ if and only if*

$$\phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{i0}}{1 - \mu_{i0} - \mu_{i1}}.$$

For the PGG, $\delta_i = \alpha_i(1 - r_i/N)$; when $\mu_{i0} = \mu_{i1}$ this reduces to $r_i > N$. A sufficient condition for $p_C^{\text{intro}} > \frac{1}{2}$ is that the inequality holds for every i .

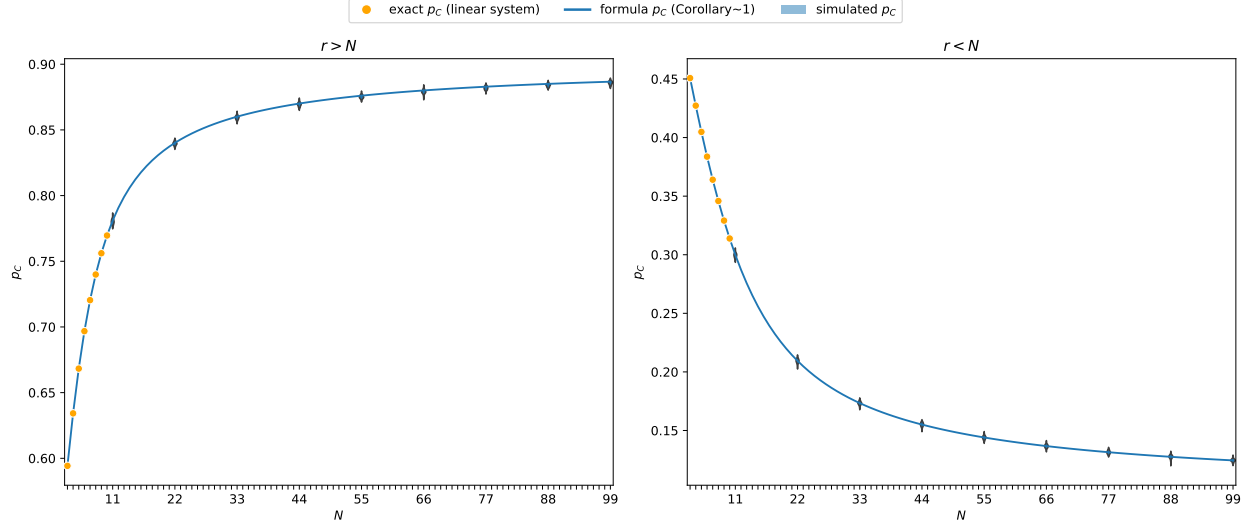


Figure 1: Validation of formula (7) on a heterogeneous group with $\alpha_i = i$, $\beta = 0.5$, $\mu_{i0} = \mu_{i1} = 0.1$. *Left:* $r = 2N$ (so $r > N$ for all N). *Right:* $r = N/2$ (so $r < N$ for all N). Orange dots: exact p_C from the full 2^N linear system (feasible only for small N). Blue curve: formula (7) (Corollary 5). Blue bands: p_C estimated from 10^5 simulation steps across 99 independent runs. All values obtained using [3].

Proof. Since $1 - \mu_{i0} - \mu_{i1} > 0$:

$$p_i > \frac{1}{2} \iff (1 - \mu_{i0} - \mu_{i1})\phi_i(\delta_i) > \frac{1}{2} - \mu_{i0} \iff \phi_i(\delta_i) > \frac{\frac{1}{2} - \mu_{i0}}{1 - \mu_{i0} - \mu_{i1}}.$$

When $\mu_{i0} = \mu_{i1}$, the right-hand side equals $\frac{1}{2}$, and since ϕ_i is strictly decreasing with $\phi_i(0) = \frac{1}{2}$, this is equivalent to $\delta_i < 0$. For the PGG, $\delta_i = \alpha_i(1 - r_i/N) < 0$ iff $r_i > N$. The claim about p_C^{intro} follows by averaging. □

Corollary 7 (Neutral drift). *If $\beta_i = 0$ for all i , then ϕ_i is constant and the payoff parameters δ_i have no effect on p_C ; cooperation is determined entirely by mutation:*

$$p_i = \frac{1 + \mu_{i0} - \mu_{i1}}{2}, \quad p_C^{\text{intro}} = \frac{1}{2} + \frac{1}{2N} \sum_{i=1}^N (\mu_{i0} - \mu_{i1}).$$

In particular, $p_C^{\text{intro}} = \frac{1}{2}$ if and only if $\sum_i (\mu_{i0} - \mu_{i1}) = 0$.

Proof. When $\beta_i = 0$, the Fermi function satisfies $\phi_i(x) = \frac{1}{2}$ for all x . Substituting into (3):

$$p_i = \frac{1 - \mu_{i0} - \mu_{i1}}{2} + \mu_{i0} = \frac{1}{2} + \frac{\mu_{i0} - \mu_{i1}}{2} = \frac{1 + \mu_{i0} - \mu_{i1}}{2}. \quad \square$$

Corollary 8 (Role of heterogeneity). *For the PGG, for player i with $r_i \neq N$:*

- (i) $\frac{\partial p_i}{\partial \alpha_i}$ is positive when $r_i > N$ and negative when $r_i < N$.

State $\mathbf{a}$	Formula $\pi_{\mathbf{a}}$	Exact $\pi_{\mathbf{a}}$
DDD	0.04193	0.04193
DDC	0.32463	0.32462
DCD	0.04193	0.04193
DCC	0.32463	0.32462
CDD	0.01526	0.01527
CDC	0.11818	0.11818
CCD	0.01526	0.01527
CCC	0.11818	0.11818

Table 1: Stationary-distribution probabilities $\pi_{\mathbf{a}}$ for all eight states $\mathbf{a} \in \{D, C\}^3$ with $N = 3$, $\alpha_i = 1$, $\beta = 2$, $\mu_{i0} = \mu_{i1} = 0.1$, and per-player multipliers $r_1 = 1 < N = r_2 = 3 < r_3 = 9$. Formula: product measure (4) (Theorem 3); Exact: values from the 2^N linear system obtained using [3]. Both columns to five decimal places.

(ii) $\frac{\partial p_i}{\partial \beta_i}$ is positive when $r_i > N$ and negative when $r_i < N$.

Proof. Write $p_i = (1 - \mu_{i0} - \mu_{i1})\phi_i(\alpha_i(1 - r_i/N)) + \mu_{i0}$.

Part (i). Differentiating with respect to α_i :

$$\frac{\partial p_i}{\partial \alpha_i} = (1 - \mu_{i0} - \mu_{i1})(1 - r_i/N) \phi'_i(\alpha_i(1 - r_i/N)).$$

Since $\phi'_i(x) < 0$ for all x and $1 - \mu_{i0} - \mu_{i1} > 0$, the sign equals $-\text{sgn}(1 - r_i/N) = \text{sgn}(r_i/N - 1)$, which is positive if and only if $r_i > N$.

Part (ii). Write $g(t) = (1 + e^t)^{-1}$, so $\phi_i(x) = g(\beta_i x)$. Differentiating with respect to β_i :

$$\frac{\partial p_i}{\partial \beta_i} = (1 - \mu_{i0} - \mu_{i1}) \alpha_i(1 - r_i/N) g'(\beta_i \alpha_i(1 - r_i/N)).$$

Since $g' < 0$ and $\alpha_i > 0$, the sign equals $-\text{sgn}(1 - r_i/N) = \text{sgn}(r_i/N - 1)$, which is positive if and only if $r_i > N$. $\square$

Remark 9. Corollaries 6–8 show that each player’s long-run cooperation is fully determined by their individual triple (α_i, β_i, r_i) and the group size N . Players with $r_i > N$ cooperate more than half the time, and increasing α_i or β_i raises their cooperation probability further; players with $r_i < N$ show the opposite sensitivity. The parameters of other players have no effect on player i ’s marginal cooperation probability.

6 Conclusion

We have proved that introspection dynamics on any state-independent game is exactly solvable. When Δf_i depends only on a_i , each selection of player i places them in state C with probability p_i regardless of current state (Theorem 2), and the stationary distribution is a product measure (Theorem 3). For the heterogeneous public goods game, the linear payoff structure implies state-independence (Lemma 4), and the long-run cooperation probability has a closed form with N terms rather than 2^N (Corollary 5).

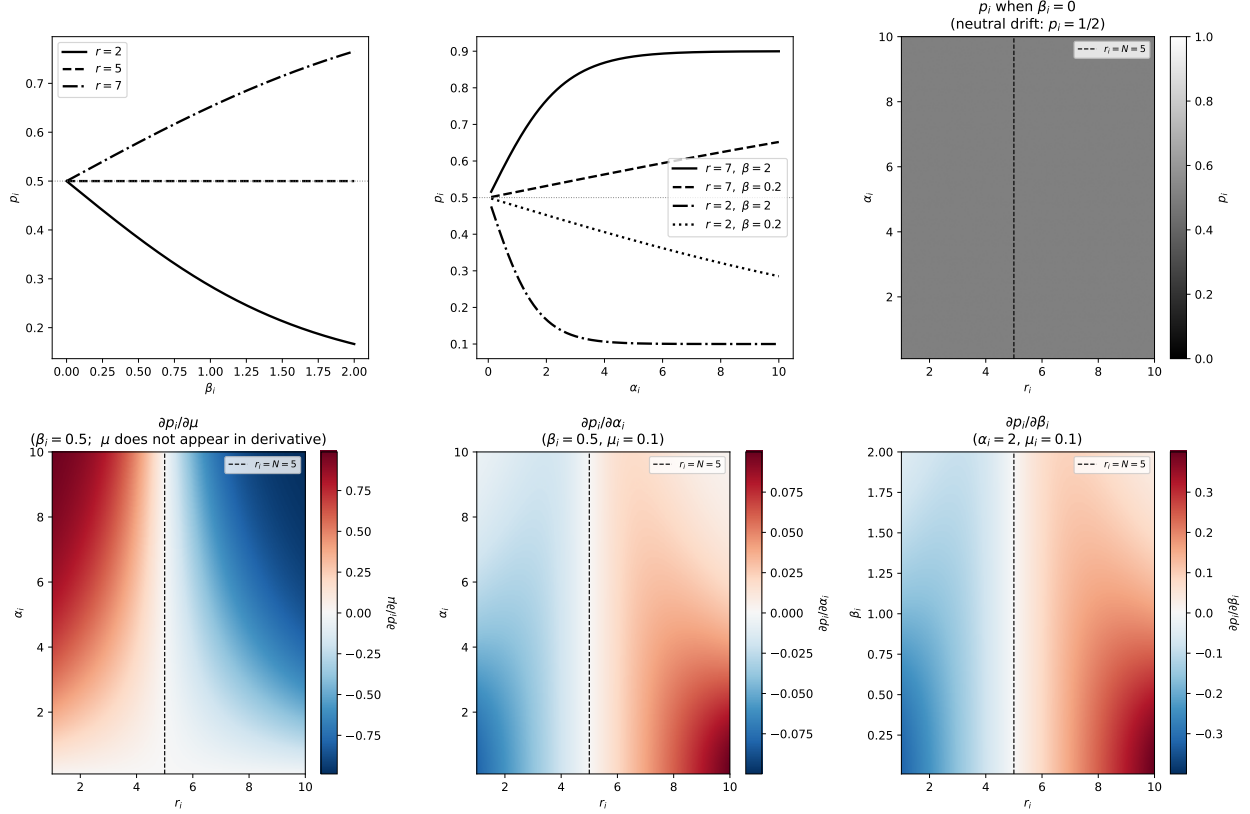


Figure 2: Structural properties of p_i for a single player with $N = 5$, $\mu_{i0} = \mu_{i1} = 0.1$. *Top left:* p_i as a function of β_i ($\alpha_i = 2$) for $r < N$, $r = N$, and $r > N$; all curves pass through $p_i = \frac{1}{2}$ at $\beta_i = 0$ (Corollary 7). *Top centre:* p_i as a function of α_i for two values of r and β_i ; curves above (below) $\frac{1}{2}$ correspond to $r > N$ ($r < N$). *Top right:* $p_i(\alpha_i, r_i)$ at $\beta_i = 0$, confirming $p_i = \frac{1}{2}$ everywhere (Corollary 7); the dashed line marks $r_i = N$. *Bottom row:* Heatmaps of the partial derivatives $\partial p_i / \partial \mu_i$, $\partial p_i / \partial \alpha_i$, and $\partial p_i / \partial \beta_i$ over the (r_i, α_i) or (r_i, β_i) plane ($\beta_i = 0.5$ or $\alpha_i = 2$ as fixed parameter). Red indicates a positive derivative; blue indicates negative. Each derivative changes sign exactly at $r_i = N$ (dashed line), consistent with Corollary 8.

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